

Body-fitted external flow validation

SIMPLE/Rhie-Chow; first order upwind; strict independent physical error <3%.

NACA reference: vector-digitized Di Ilio 2020, uncertainty +/-0.0001.

Historic NACA 0.205/0.120 pass claim is withdrawn.

Single grid and residual convergence do not establish physical accuracy.

naca 64x26 cpu: converged=True, passed=False

{'drag': 0.12097920058753613, 'lift': 0.20622549060809398}

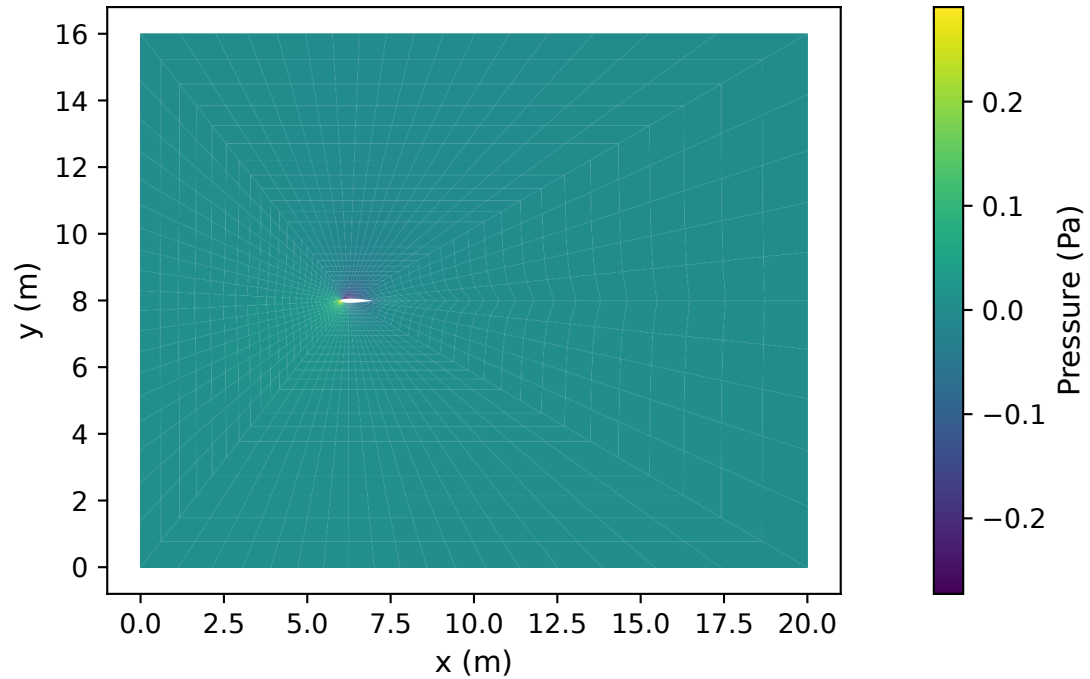
{'drag_relative_error': 0.03299064480299398, 'lift_relative_error': 0.02669417201308529, 'solver_residual': 9.940574969947795e-07, 'convergence_failure': 0}

naca 128x52 cpu: converged=True, passed=False

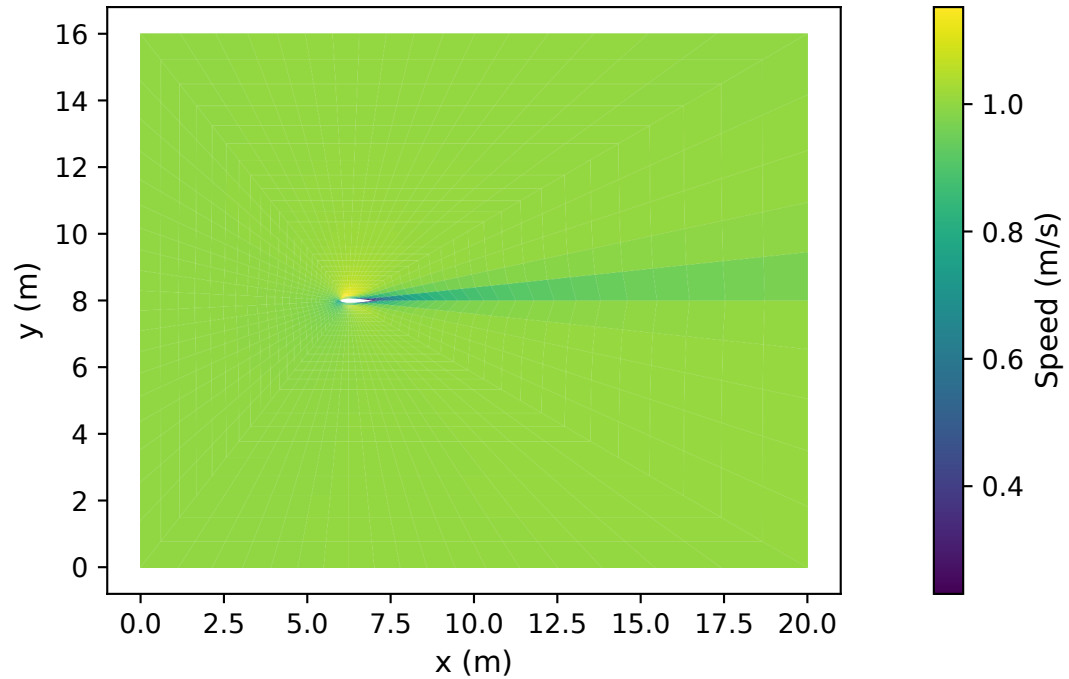
{'drag': 0.14878141027429978, 'lift': 0.24299637845736927}

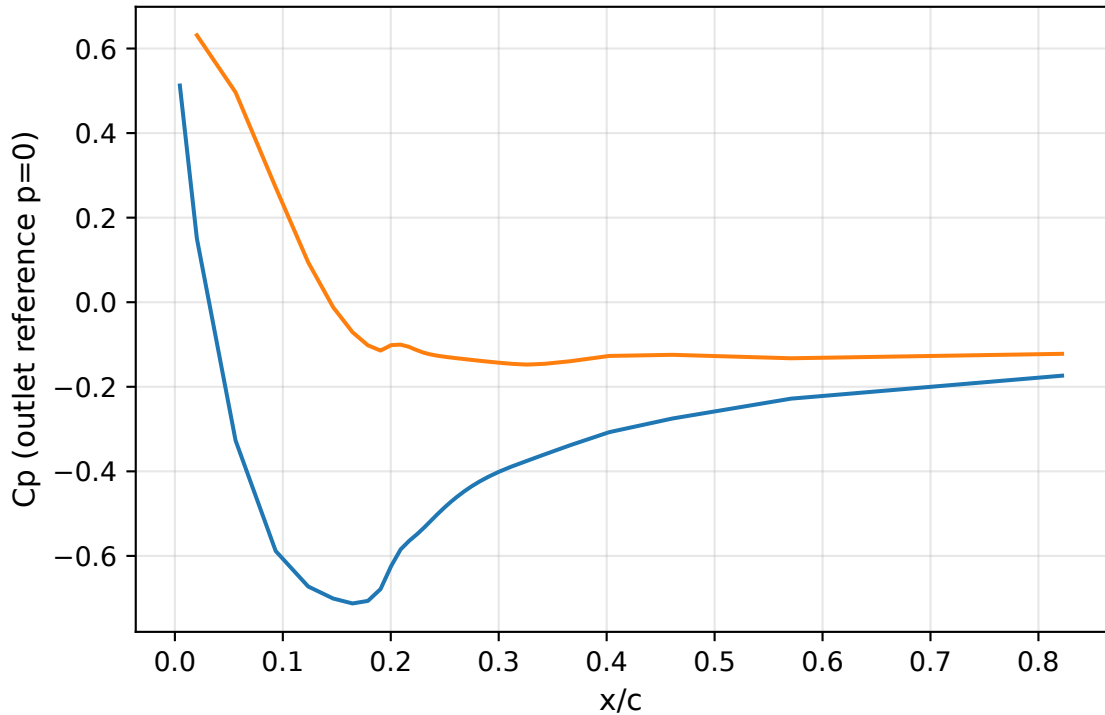
{'drag_relative_error': 0.19114181587145385, 'lift_relative_error': 0.20975813827291873, 'solver_residual': 9.887651046099656e-07, 'convergence_failure': 0}

naca 64x26

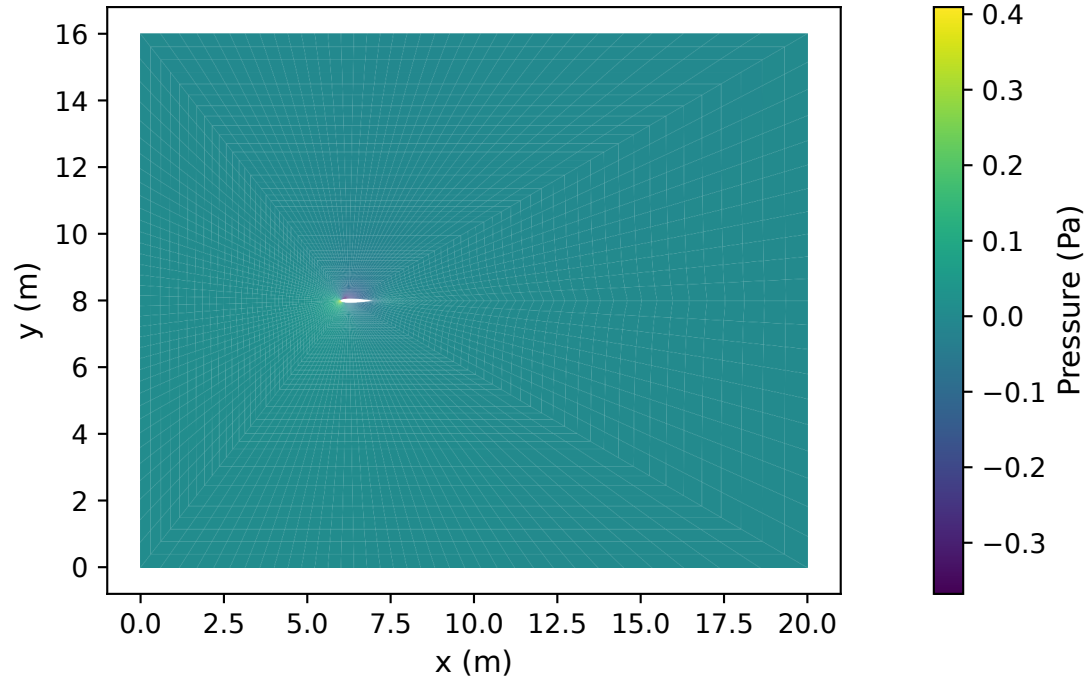


naca 64x26





naca 128x52



naca 128x52

